



Date: 14-01-2026	CIE 3	Max. Marks: 10+50
Semester: VII	UG	Duration: 2 Hrs (1:20PM- 3:20PM)
Course Title: Mathematics of Music		Course code: MA375TGR

Department of Mathematics

S No	Quiz	M	BT	CO
1.	Define avartana and sama in the context of the Carnatic tala system.	2	1	2
2.	State the tihai formula $3P + 2G = T$ and explain what each variable represents.	2	2	1
3.	What is a fractal? Give one example of a deterministic fractal.	2	2	2
4.	Define critical band in psychoacoustics and state its significance for consonance perception.	2	3	3
5.	Calculate the total number of aksharas in Misra-jati Rupaka Tala.	2	2	3

S No	Test	M	BT	CO
1a	Explain the hierarchical organization of time in Carnatic music: matra, akshara, and avartana. How do these relate mathematically through jati and nadai?	5	2	1
1b	A tala has structure I O I I with Khanda jati. Calculate the number of aksharas in one cycle. If rendered in Misra nadai, how many matras are in one avartana?	5	2	1
2a	Design a tihai for 2 cycles of Adi Tala in Chatusra nadai (64 matras). Specify the pattern length P and gap G using the tihai formula $3P + 2G = T$. Verify that your design ends at sama. (ii)	5	3	2
2b	A korvai has purvanga of 12 matras and uttaranga of 8 matras. If repeated 3 times, calculate the total length. Does it fit exactly in 2 cycles of Rupaka Tala (Chatusra jati, Chatusra nadai)? Justify.	5	3	2
3a	Explain the concept of 1/f noise (pink noise) in music as discovered by Voss and Clarke. How does it relate to entropy and redundancy in musical composition?	5	3	4
3b	Calculate the Shannon entropy and redundancy for a four-note melody system with probabilities: Sa (0.4), Ri (0.3), Ga (0.2), Ma (0.1). Interpret your results in terms of listener engagement.	5	3	4



4	(i) Define critical bandwidth using Zwicker's formula. Calculate the critical bandwidth at 520 Hz. (ii) Using your result from (i), analyze whether an equal temperament minor third interval (520 Hz and _____ Hz) would be perceived as consonant or dissonant. Explain the relationship between critical bands and the perception of roughness.	10	2	3
5a	Define a first-order Markov chain. Construct a valid 3×3 transition matrix for generating melodies using notes {C, D, E} where stepwise motion has probability 0.5, staying on the same note has probability 0.3, and skipping has probability 0.2. Verify your matrix is row-stochastic.	5	4	4
5b	Explain the missing fundamental phenomenon in psychoacoustics. A complex tone contains harmonics at 200 Hz, 300 Hz, 400 Hz, and 500 Hz but lacks the fundamental. What pitch will a listener perceive and why?	5	4	4